

ADS - Week - 1

L1 Intro to ADS

- we know already few algs like PCA / SVD, basic classifiers (NN, NB, log Reg, Trees)
- we are looking at algorithms on aspects of few things like scale, graphs
- prerequisites $\rightarrow$ probability, linear algebra, optimization
- we are going to use "randomiza" as a tool to deal with scale $\Rightarrow$ we will understand this better over the course.

L2 Concentration Inequalities: Markov's Inequality

- some random experiments yields a RV.
die eg. $\rightarrow X: SS = \{1, 2, 3, 4, 5, 6\}$
event space $\Rightarrow$ power set of $\{1, \dots, 6\}$
probab. space $\rightarrow \left\{ \frac{1}{6}, \frac{1}{6}, \dots, \frac{1}{6} \right\}$
- but what the case where the entire probab space for a random variable is not known, how do we still make meaningful insights?
 $\Rightarrow$ we start with conc. inequality

- Markov Inequality, $\Rightarrow$

For any non-negative RV x & any $a > 0$

$$P(x \geq a) \leq \frac{E[x]}{a}$$

$\rightarrow E[x]$

not accurate, but still gives an upper bound

what is the chance that this expt. will lead to a value, greater than or equal to a ?

$$\text{eg. } P(x \geq 5) \leq \frac{35}{10 \times 5} \Rightarrow \leq 0.7$$

0.33

this is at most possible.

we could say, most trivial upper bound is 1.

eg. let's say $x \rightarrow$ dist. of a data pt. to cluster center in k -means.

$$E[x] = 5$$

if for a randomly picked pt., what is the probab. that dist. to cluster center is ≥ 20 ?

$$P(x \geq 20) \leq \frac{5}{20} \leq 0.25$$

L3 Chebyshev's Inequality

- there can be RV where means are same but variance could be different for them.

For any RV X with finite $E[X]$ and finite variance σ^2 , for any $k > 0$:

$$P(|X - E[X]| \geq k\sigma) \leq \frac{1}{k^2}$$

how much the RV
deviates from the mean
in absolute terms

is
more
than k times S.D.

→ larger deviation means lesser events.

L4 Hoeffding's Inequality

Let $X_1, X_2, \dots, X_n$ be independent RV where each $X_i \in [a_i, b_i]$

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$$

$$P(\bar{X} - E[\bar{X}] \geq t) \leq e^{-\frac{2nt^2}{(b-a)^2}}$$

for the coin toss ex...

$$P(\bar{X} - p \geq t) \leq e^{-2nt^2}$$

probab. of the sample mean being far from the expected means drops exponentially if $n \uparrow$ or deviatⁿ t increases.

L5

Useful Inequalities in Probab.

- Union bound,

$$P(A \cup B) = P(A) + P(B) - \overset{\rightarrow \geq 0}{P(A \cap B)}$$

this is difficult

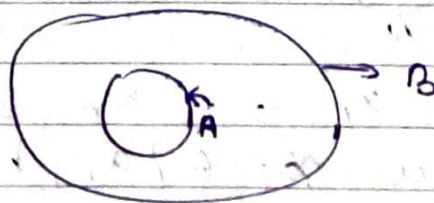
$$\boxed{P(A \cup B) \leq P(A) + P(B)}$$

$$P(A_1 \cup A_2 \cup \dots \cup A_n) \leq \sum_{i=1}^n P(A_i)$$

-

$$A \Rightarrow B$$

if A implies B



$$\boxed{P(A) \leq P(B)}$$

1.6 Johnson-Lindenstrauss (JL) Lemma: Intro

- Given n pts in a high-dimensional space $\mathbb{R}^d$ & an error parameter $\epsilon \in (0, 1)$, then there exists a mapping $F: \mathbb{R}^d \rightarrow \mathbb{R}^k$

with $k \in O\left(\frac{\log n}{\epsilon^2}\right)$, st. $\forall x, y$

$$(1 - \epsilon) \|x - y\|_2^2 \leq \|F(x) - F(y)\|_2^2 \leq (1 + \epsilon) \|x - y\|_2^2$$

dist. in the k space is not too far away as dist. in original space.

eg. $n = 10^6$, $d = 10,000 \rightarrow k = O\left(\frac{\log 10^6}{\epsilon^2}\right)$

$\epsilon = 0.1$

$k \approx 2000$

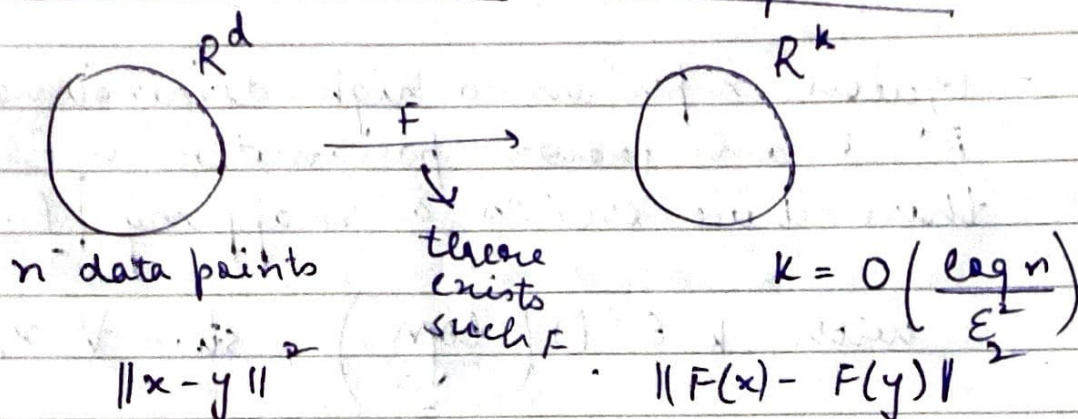
$$0.9 \|x - y\|_2^2 \leq \|F(x) - F(y)\|_2^2 \leq 1.1 \|x - y\|_2^2$$

$\downarrow \quad \quad \quad \uparrow \quad \quad \quad \uparrow \quad \quad \quad \downarrow$
 $\mathbb{R}^{10,000} \quad \quad \quad \mathbb{R}^{2000} \quad \quad \quad \mathbb{R}^{2000} \quad \quad \quad \mathbb{R}^{10,000}$

- the best thing about this lemma is during dim. reducⁿ, it has nothing to do with 'd' (original dim.), it only depends on no. of data points and how much error, I am willing to tolerate.

L7

JL Lemma - Part 1 - No Isometry Exists



Lemma

$$(1-\epsilon) \|x-y\|^2 \leq \|F(x)-F(y)\|^2 \leq (1+\epsilon) \|x-y\|^2$$

- Linear transformation $\mathbb{R}^d \rightarrow \mathbb{R}^k$

$$u \xrightarrow{\mathbb{R}^d} Ru \xrightarrow{\mathbb{R}^k}$$

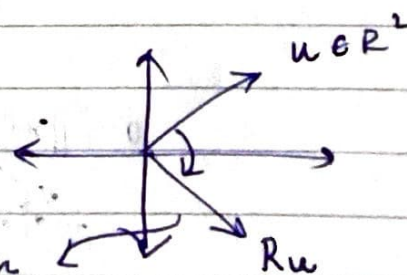
$$\|u\|^2 = \|Ru\|^2 \quad (\text{if we are greedy})$$

$$= (Ru)^T (Ru)$$

$$u^T u = u^T (R^T R) u$$

then this has to be a identity matrix

$$\begin{matrix} (d \times k) & (k \times d) \\ \downarrow & \\ d \times d \end{matrix}$$



len does not change after rotation (linear op)

- then only this property will be preserved.

then $\{a_1, \dots, a_d\}$ all of them should be orthonormal vectors

→ we can only find at max k orthonormal vectors and not d vectors.

$$\|u\| \neq \|Ru\|^2$$

isometry does not exist

↳ we get approximate isometry because of epsilon parameter.

- one more thing, we have a relaxⁿ that we don't have to save everything present in d -dimensional space, we just have to remember the n -pairwise comparisons.
- we try to get a randomised matrix, which when applied to a vector $\Rightarrow$ it atleast preserves the notation.

L8

Part 2: Approximate Isometry

Goal: is to now approx. isometry (which is the JL guarantee) for pairwise distances of any n -points in $\mathbb{R}^d$.

$$R = \begin{bmatrix} a_1^T \\ a_2^T \\ \vdots \end{bmatrix} \begin{bmatrix} | \\ u \\ | \end{bmatrix}_{\mathbb{R}^d} = \begin{bmatrix} a_1^T u \\ a_2^T u \\ \vdots \end{bmatrix}_{\mathbb{R}^k}$$

$$R = [a_{ij}]$$

$$a_{ij} \sim N(0, 1)$$

comes from std.
gaussian distribⁿ

- let's examine $x_i^T u$

$\leftarrow d\text{-dim}$

u_i

$\downarrow$
 $N(0, 1)$

$$x_i^T u = \sum_{i=1}^d x_{i,i} u_i$$

this is also random

$$E[x_i^T u] = E\left[\sum_{i=1}^d x_{i,i} u_i\right]$$

$$= \sum_{i=1}^d \underbrace{E[x_{i,i}]}_{0 \forall i} u_i$$

$$E[x_i^T u] = 0$$

- what abt its variance?

$$\text{var}(x) = E(x^2) - (E[x])^2$$

$$\text{var}(x_i^T u) = E[(x_i^T u)^2] - \underbrace{(E[x_i^T u])^2}_0$$

$$= E\left[\left(\sum_{i=1}^d x_{i,i} u_i\right)^2\right]$$

$$= E\left[\sum_{i=1}^d (x_{i,i} u_i)^2 + 2 \sum_{\substack{j,k \\ j \neq k}} x_{i,j} x_{i,k} u_j u_k\right]$$

$$= \sum_{i=1}^d E[u_i]^2 u_i^2 +$$

$$2 \sum_{\substack{j,k \\ j \neq k}} \underbrace{E[x_{i,j} x_{i,k}]}_{E[x_{i,j}] E[x_{i,k}]} u_j u_k$$

$$\begin{aligned} \text{var}(u^T u) &= \sum_{i=1}^d \text{var}(u_i) u_i^2 \\ &= \sum_{i=1}^d u_i^2 = \|u\|^2 = \text{var}(u^T u) \end{aligned}$$

$$\begin{bmatrix} u_1^T u \\ u_2^T u \\ \vdots \\ u_k^T u \end{bmatrix} \rightarrow N(0, \|u\|^2)$$

$$\text{best guess for } \|u\|^2 = \frac{\sum_{i=1}^k (u_i^T u - 0)^2}{k}$$

Transformation

$$M = \frac{R}{\sqrt{k}}$$

$$u \in \mathbb{R}^d \rightarrow Mu \in \mathbb{R}^k$$

$$\|Mu\|^2 = \begin{bmatrix} -u_1^T u / \sqrt{k} \\ -u_2^T u / \sqrt{k} \\ \vdots \end{bmatrix}^2 = \frac{\sum_{i=1}^k (u_i^T u)^2}{k}$$

L9 Goodness of Estimate : Part - 3

$$u \in \mathbb{R}^n \xrightarrow{M = \frac{1}{\sqrt{k}} R} Mu \in \mathbb{R}^k$$

$N(0, 1)$

$\|Mu\|^2 \Rightarrow$ is a good estimator for $\|u\|^2$ (unbiased)

Now, the pt. is that how good is this estimator.

same. $u \in \mathbb{R}^d$ $v = \frac{1}{\sqrt{k}} R u$ $v \in \mathbb{R}^k$

$$v_i = \frac{1}{\sqrt{k}} \sum_{j=1}^d r_{ij} u_j$$

$E[\|v\|^2] = \|u\|^2 \Rightarrow$ unbiased nature
if we perform the expt. infinite no. of times, then we can preserve the pairwise distances.

if I do this process only once, then

$$P\left(\underbrace{\|v\|^2}_{\mathbb{R}^k} \geq \underbrace{(1+\epsilon)\|u\|^2}_{\mathbb{R}^d}\right)$$

this a bad event as it is not preserving any info.

define $x = \frac{\sqrt{k} v}{\|u\|} \rightarrow \mathbb{R}^k$ $v = \frac{1}{\sqrt{k}} R u$

$$\|x\|^2 = \left\| \frac{\sqrt{k} v}{\|u\|} \right\|^2 = \frac{k \|v\|^2}{\|u\|^2}$$

$$E[x] = E\left[\frac{\sqrt{k}}{\|u\|} \frac{1}{\sqrt{k}} u^T u \right] = E\left[\frac{u^T u}{\|u\|} \right] = 0$$

$$(\text{cov}(x))_{ii} = \frac{E((u_i^T u)^2)}{\|u\|^2} = 1$$

$$\begin{bmatrix} 1 & & 0 \\ & \ddots & \\ 0 & & 1 \end{bmatrix} (\text{cov}(x))_{ij} = E \left[\frac{\sqrt{k}}{\|u\|} \frac{1}{\sqrt{k}} u_i^T u \right]$$

$$= \frac{1}{\|u\|^2} E[(u_i^T u)(u_j^T u)]$$

identity matrix in k dimensions = 0

$$u \xrightarrow{\frac{1}{\sqrt{k}} R} v = \frac{1}{\sqrt{k}} R u \rightarrow x = \frac{\sqrt{k}}{\|u\|} v$$

$$E[x] = 0, \text{cov}(x) = I_{k \times k}$$

x is a std. multivariate normal var.

$$x \sim N(\vec{0}, I_{k \times k})$$

$$\|x\|^2 = \frac{k \|u\|^2}{\|u\|^2}$$

$$P(\|v\|^2 \geq (1+\epsilon) \|u\|^2) \sim \text{this is wanted}$$

$$= P(\|x\|^2 \geq (1+\epsilon) k)$$